what it tells us about fundamental physics and our cosmological model. But *I* know it's really the dread.

6 Yes, we do this. It's called laser ranging, and the only reason we CAN do it is because the Apollo astronauts left a mirror up there. It's a handy tool for both seeing how far away the Moon is (fun fact: it's drifting away from the Earth at almost four centimeters per year) but also for testing how gravity works, by watching the orbit very, very carefully.

7 She wasn't referred to as an astronomer at the time. She was one of a group of women called "computers," who were hired to examine astronomical plates as cheap labor and who ended up doing a huge number of foundational calculations in astrophysics. Edwin Hubble, who used her discovery to measure the size and expansion of the universe, later said she deserved a Nobel Prize. Unfortunately, beyond being known and respected by her immediate colleagues, she was almost entirely unacknowledged in her lifetime.

8 I like to think of the bright Cepheids like giant lazy Saint Bernard dogs, while the dim ones are excitable jumpy Chihuahuas.

9 Simple for stars, anyway.

10 Based on current estimates, the Sun is already increasing in radius at about an inch per year. But at the same time, the Earth is expanding its orbit, such that we're moving away from the Sun at about 15 centimeters per year (I make no apology for the mixed units here), so it's not like the surface of the Sun is getting closer to us at the moment.

11 Supernova 1006, seen between April 30 and May 1, 1006, was likely a Type Ia supernova caused by the collision of two white dwarf stars about 7,000 light-years away, in our own galaxy. The remnant, which in astronomical images looks a lot like a colorful ball of smoke, can still be seen today.

12 They're "dwarfs," not "dwarves," for reasons that are not entirely clear.